Ming-Kai Kan

Robotics Control & Software Engineer | Factory AMR Navigation & Production ADAS Trajectory Planning

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PROFESSIONAL SUMMARY

- **8 years in control system development, 6 of them on production programs, robotics control and vehicle control:** motor control tuning for PMSM actuators, lateral trajectory planning and control on production vehicles, and Isaac Sim Sim-to-Real for factory AMRs running on live production lines.
- **Servo and actuator control:** tuned the field-oriented current loop controller parameters on a TI C2000 DSP using the id / iq step response as the criterion, identified actuator dynamics, and ran recursive least squares at a 10 ms cycle on the vehicle. Found a Simulink code generation defect during implementation and, after confirming it with the MATLAB distributor, rewrote the control computation in fixed-point Q format.
- **Sim-to-Real and ROS 2 deployment:** the native Nav2 global planner still could not fit the unstructured plant floor after parameter tuning, so it was replaced with an in-house graph-node global planner. Built an unattended overnight regression test in Isaac Sim covering 300 goal points; station task success rate rose from 40% to 99% on the same baseline batch, after which test difficulty was progressively tightened, and 3 robots were deployed on the plant floor.
- **Vehicle control and trajectory planning architecture:** led the convergence of the ADAS lateral trajectory planning architecture, unifying planning and control modules that had been implemented separately and redesigning them into a low-complexity, low-computation, highly reusable shared module that meets ISO 26262 requirements for verifiability and complexity control. 9 lateral features converged into a single shared module; TC277 computation load dropped from 100% to 37%; new feature development went from 4 months to 2–3 days and validation from 1 year to 1 month.
- **Functional safety implementation:** carried ISO 26262 from standard clauses through to production code, establishing complete requirement traceability and a verification evidence chain across 2 production modules. Delivered to 4 OEM customers and 5 production vehicle programs.
- **Turning production-line engineering problems into patents:** 4 invention patents filed, 3 granted; originator of the concept on 2 of them (1 as sole inventor, 1 as first inventor).

PROFESSIONAL ATTRIBUTES

- **Able to reframe the problem:** a lane-line far-range jitter issue that the perception supplier assessed as requiring a new module was instead handled downstream in the ADAS system through polynomial order reduction, saving a six-figure USD redevelopment cost and the associated schedule at close to zero cost.
- **Not bound by domain convention:** automotive lateral trajectory planning has long been dominated by MPPI and MPC, which are computationally heavy and hard to keep behaviorally consistent in edge scenarios. I brought trajectory planning logic from robotic arms into ADAS, bringing computation load down into a stable operating range, making steering control behavior more consistent, and allowing one set of logic to cover every lateral feature.
- **Strong system architecture evolution and refactoring capability:** facing differing customer requirements, the priority is to decompose them by what they have in common and then decide whether to absorb them into the existing shared module or restructure the architecture, instead of continuing to patch. 9 lateral features across the ADAS system were consolidated into a single shared module; development and validation schedules compressed together, and one defect fix benefits the entire system.
- **Effective at visual collaboration and cross-functional communication:** real-time visualization of algorithm internal state, control output and sensor data, including LiDAR point cloud overlays, live map rendering, path planning and state machine transitions. Debugging does not rely on tuning parameters by feel; most issues are located the moment they appear on screen, and in cross-functional discussions the other side can see what is being argued about.

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB / Simulink
- **Embedded and Real-Time Systems:** Model-based development with automatic Simulink code generation to embedded C (TI C2000 F28069, Infineon AURIX TC277 / TC397); fixed-point Q format implementation; bare-metal timer-interrupt control loops; loop timing and jitter analysis; reading generated C code for target-board debugging
- **Robotics and Middleware:** ROS 2 (Humble/Jazzy), Nav2, DDS, rclpy, node lifecycle management, system design specification writing
- **Motion Control and Estimation:** Global and local path planning, polynomial trajectory generation, actuator dynamics modeling and system identification, field-oriented current loop PI tuning in dq coordinates, position and force closed-loop control, sensor calibration and bias compensation, SLAM and relocalization, Kalman filter and state estimation, recursive least squares, fuzzy control
- **Simulation and Validation:** Isaac Sim, Sim-to-Real transfer, unattended overnight regression testing, dSPACE rapid prototyping, MIL / SIL / CIL / HIL / FOT, Vector CANalyzer, INCA
- **Platforms and Buses:** Jetson (Orin / Thor), Infineon TC277 / TC397, TI C2000 F28069, CAN / CAN FD (DBC signal and message design), RS485 (latency and cycle jitter diagnosis for control loops), control circuit design
- **Deployment:** Linux, Docker, GitLab CI/CD (self-hosted ARM Runner on target hardware), Git
- **Processes and Standards:** ISO 26262, ASPICE, V-Model, Polarion, requirement traceability and verification evidence chain

WORK EXPERIENCE

2025.05 ~ Present	**FIH · Robotics (Foxconn Group)** **Senior Engineer · Factory AMR Navigation**

Company and Project:

- Robotics business unit of the Foxconn Group, developing autonomous mobile robots (AMRs) for the group's own plants.
- The operating environment is closer to a semiconductor fab than to a warehouse: mixed human and robot traffic, no fixed guide paths, continuous shift operation, and station docking repeatability as the acceptance criterion.

Responsibilities:

- Own the plant AMR navigation software: global path planning, simulation validation framework, CI pipeline.
- **Established an end-to-end virtual-to-physical development flow:** led the full cycle from algorithm development through simulation validation to on-robot deployment, with 70–80% of my time spent on hands-on implementation.
- Collaborate across 3 groups (brain / upper body / fleet management system) and define the interface contracts between them.

Key Contributions:

- **In-house development and replacement of core navigation modules:** applying the same approach used for ADAS lateral architecture convergence, a single architecture supports both mapped and unmapped modes, covering station-topology navigation and free navigation. Mapped areas use station-topology planning with map-based localization; unmapped areas switch to local cost-map planning with obstacle avoidance, and localization switches to a mode that does not depend on a prior map. The robot can operate both in unstructured plant areas and where the map does not yet reach.
- **Ability to establish Sim-to-Real convergence methods:** built an unattended Isaac Sim test framework that batch-generates 300 station tasks, automatically classifies failure causes and produces success-rate reports, letting the algorithm iterate once per day. A graphical interface renders control output in real time, making the gap between simulation and the real robot visible on screen. On the same baseline batch, station task success rate rose from 40% to 99% within one month (297/300; the remaining 3 only missed the time threshold, with correct paths), and 3 robots were deployed on the plant floor.
- **Ability to build target-hardware validation flows:** split into L1 unit / L2 integration / L3 system tiers following a V-Model layered validation architecture; Runners are deployed on Jetson, so L1 and L2 run directly on the ARM target hardware instead of being simulated on an x86 development machine. Platform-dependent behavioral differences and resource limits are caught at commit time and do not accumulate until on-robot integration.
- **Replacing manually maintained cross-group interface management with automation:** CI automatically scans and produces a system-wide list of topic / message definitions, data formats and publish-subscribe relationships, updated daily. Cross-group interfaces use this list as the single source of truth, replacing manually maintained interface documents. Architecture diagrams are likewise kept in Mermaid in the same repository as the code and versioned alongside it, so the validation team confirms the actual control flow from the diagram instead of tracing source line by line.
- **Skills:** ROS 2 / Nav2, Isaac Sim, Python, SLAM, Motion Planning, Jetson (Orin / Thor), Linux, GitLab CI/CD, Docker

2021.06 ~ 2025.06	**MobileDrive (Foxconn Group × Stellantis joint venture)** **Senior Engineer · ADAS · Technical Lead, Traffic Jam Assist (TJA) System**

Company and Project:

- ADAS software joint venture between Foxconn and Stellantis, delivering L2 production driver assistance systems to domestic and international OEM customers, including BYD, CMC (China Motor Corporation), Lexgen and Windrose.

Responsibilities:

- Technical lead for the lateral system, covering map data fusion, lane-line perception fusion, trajectory planning and lateral control, from architecture design and algorithm implementation through production delivery to automotive process adoption.
- Led the convergence of the lateral trajectory planning architecture and gave technical direction on architecture decisions and interface contracts to each feature development team.
- Primary technical point of contact for OEM customers, perception suppliers and map data suppliers.

Key Contributions:

- **System-level architecture convergence and refactoring capability:** identified the root cause of each lateral feature implementing its own trajectory logic, abstracted every lateral maneuver in the system into two prototypes, “along the reference line” and “relative offset”, and built a three-segment trajectory generator from fourth-order polynomials, unifying the team's previously separate planning modules. 9 lateral features across the system converged into a single shared module, 6 of which I delivered independently and 3 of which colleagues integrated in about 2 days each; TC277 computation load dropped to 37%, new feature development went from 4 months to 2–3 days and validation from 1 year to 1 month.
- **Hands-on experience implementing functional safety requirements:** carried ISO 26262 from standard clauses through to production code, establishing requirement traceability, safety analysis and a verification evidence chain in the LFS and TRJ production modules. The company obtained DEKRA ISO 26262 ASIL-D certification during the same period (full ADAS system scope, a team achievement).
- **Extensive direct customer-facing experience (on-site and remote):** independently handled a Chinese map data supplier and completed automated highway on-ramp and off-ramp maneuvers using lane-line perception in place of HD maps, with 2 cm in-lane lateral positioning accuracy. Demonstrated L2.9 highway NOA on site to 6 OEMs and Tier 1 suppliers.
- **L2 production delivery to 4 OEM customers and 5 production vehicle programs (3 in Taiwan, 2 in China).**
- **Skills:** MATLAB / Simulink, Python, TC277 / TC397, CAN / CAN FD, Vector CANalyzer, INCA, dSPACE, Polarion

Reason for Leaving: internal transfer within the group to FIH, taking on the system-level lead role for lateral control software.

2020.05 ~ 2021.06	**FIH · ADAS (Foxconn Group)** **Engineer · Lane Fusion System (LFS)**

Company and Project:

- ADAS business unit of FIH (Foxconn Group), developing lane fusion software for L2 production programs delivered to OEM customers.

Responsibilities:

- Multi-source fusion of lane lines and objects, from development through to L2 production release.
- Analyzed and modeled the output characteristics of 4 lane-line perception suppliers, quantifying their failure boundaries and parameter characteristics as the basis for platform selection and system-side compensation design.
- Built a scenario reconstruction mechanism able to restore the scene at the moment of an ADAS failure and run regression validation.

Key Contributions:

- **Ability to deploy production perception bias compensation:** developed the C1 Bias Estimator, which uses observations taken while driving to estimate the residual camera mounting angle bias left after end-of-line calibration, turning a one-time production-line calibration tool into resident production logic that corrects online. Deployed on 3 production vehicle models, correcting 86.81% of the residual bias, with zero customer complaints since launch.
- **Strength in solving supplier-level problems from the system side:** diagnosed that C0–C3 lane lines swing beyond the effective viewing range because the higher-order terms diverge, which is a problem in the output representation layer, and handled it on the system side with order-reduced refitting plus continuous weighted blending, avoiding a supplier redevelopment quotation (USD 470–625K) at zero hardware cost. Also extended scenario determination with pre-curve recognition and a timer mechanism, resolving the distortion where a curve at short viewing range cannot be represented by C0–C1. The approach was designated by management as the standard production solution.
- **Technical lead experience co-developing with an international Tier 1:** co-developed the lane fusion module with Siemens under ISO 26262, fusing lane-line detection with ego vehicle dynamics using a Kalman filter, and carried it from development through to production on 2 vehicle models.
- **Long-distance road validation and supplier evaluation experience:** ran a 1,600 km Mini-FOT from Shenzhen to Shanghai, establishing a lane-line labeling process and a scenario replay mechanism able to reconstruct the scene at the moment of failure and run lateral regression tests; completed a comparative evaluation of three perception suppliers, Mobileye, StradVision and Horizon, as the basis for platform selection. The perception parameter characteristics and failure boundaries derived from road testing were then fed back as labeling and parameter guidance for training data, supporting the launch of the company's in-house lane detection system.
- **Skills:** MATLAB / Simulink, Python, Kalman Filter, CAN, camera perception interface, data logging and scenario replay

Reason for Leaving: internal transfer within the group to MobileDrive, taking on the system-level lead role for lateral control software.

2019.01 ~ 2020.10	**NCSIST Industry-Academia Collaboration Project (concurrent with master's program)** **Lead Engineer · Motorcycle ABS**

Company and Project:

- Contract project building a motorcycle ABS control system on NCSIST proportional-valve hydraulic hardware, connecting that hardware with my advisor's control theory, carried out alongside my master's program.

Responsibilities:

- Stationed at NCSIST as the technical point of contact for the collaboration, reporting progress weekly and arranging test acceptance, building the control system from scratch around the hardware constraints, and leading a 4-person cross-disciplinary team.

Key Contributions:

- **Integration and validation capability from rapid prototype to vehicle:** used dSPACE rapid prototyping to connect the control algorithm, drive circuit, proportional-valve hydraulics and the vehicle, incorporating actuator response lag and nonlinearity into the controller design. Vehicle braking distance was 20% shorter than a conventional system, close to the theoretical minimum.
- **Strength in designing estimators from existing sensors:** designed two estimators, for vehicle speed and for peak road friction coefficient, using only the sensors already present on a commercial motorcycle with no added hardware; vehicle speed estimation error was 1.2%–2.3%. The friction coefficient estimation method was granted patent TWI719598B, on which I am first inventor among a team of 6.
- **Skills:** MATLAB / Simulink, dSPACE rapid prototyping, control circuit design, fuzzy control, parameter estimation, MIL → RCP → vehicle testing

2016.09 ~ 2019.06	**National Taipei University of Technology, Vehicle Control Laboratory (ITRI and Acer contract projects)** **Research Engineer · Brake-by-Wire Actuator Control**

Company and Project:

- Brake-by-wire actuation layer development (2 formal ITRI research contracts, 1 Acer autonomous driving project): the gap common to all three was the absence of a brake-by-wire system able to control hydraulic pressure precisely and reach a specified vehicle deceleration. Joined in the fourth undergraduate year through early admission and continued into the master's program.
- Signal-level characterization and control interface reconstruction of the iBooster and EHB: with no OEM specification or control documentation available, bench measurements established the relationship between actuation characteristics and control signals, from which a usable pressure control interface was derived as the design basis for the in-house actuation layer.
- Results deployed in the brake actuation layer of the ITRI CPEV and the Acer test vehicle: the booster alone can build pressure from 0 to 17 bar within 0.32 s, producing roughly 0.5 g of deceleration; the hydraulic module can regulate pressure on each of the four wheels independently up to 83 bar. Both expose a CAN interface so partners can issue deceleration commands directly, supporting their autonomous test vehicle development.

Responsibilities:

- Principal engineer for the project: reported progress and deliverables to ITRI and Acer, set the technical direction for brake control, and implemented the controller and hardware on the vehicle.

- Worked in a two-person team with a PhD student to reverse-engineer two production brake actuators (Bosch i-Booster electromechanical booster, BWI electro-hydraulic module), replacing their OEM controllers with in-house electronic circuits and control loops running on a TI C2000 F28069 DSP.
- My scope: current loop tuning and validation, system identification, actuator modeling, force estimation, and the hydraulic module control strategy. Motor drive hardware and FOC implementation were handled by the PhD student.

Key Contributions:

- **Experience tuning motor current control loops:** starting from three-phase PWM and DC bus measurements, tuned the PI gains of the field-oriented current controller on the DSP, judging convergence from the id / iq step response in rotating coordinates instead of visible motor motion; applied standard d-q axis decoupling until iq converged to the value corresponding to static thrust.
- **Toolchain-level debugging capability:** Simulink generated code behaved abnormally on the F28069; investigation confirmed a code generation defect, not a tuning issue. The issue was raised with MathWorks through the local distributor and confirmed by the vendor, after which the affected control computation was rewritten in fixed-point Q format.
- **Deep expertise in system identification and modeling:** on an instrumented test bench whose hydraulic topology matched the vehicle, swept the return spring with a tensile testing machine at five speeds and characterized it as three linear segments plus constant Coulomb friction, with residuals within a 3 kg band; identified the motor thrust constant and the 6.66 mm full stroke; and established the actuator force balance equation underlying the stroke-to-pressure control strategy.
- **Strength in force estimation using production sensors:** derived a Driver Force Estimator from the difference between the pedal stroke sensor and the motor absolute position, distinguishing whether the driver is pressing the brake or the motor is pushing the pedal; rewrote an offline least-squares fit as recursive least squares running at a 10 ms cycle on the vehicle. Also calibrated the mounting bias of the AS5140H absolute angle sensor, which is the reference for the entire stroke-to-force conversion.
- **Experience with hydraulic module control:** reduced pressure build, hold and release to three energization combinations of the same four solenoid valves plus a pump duty cycle, switched at millisecond scale, so a smooth pressure curve comes from mode switching, not from holding a valve open for long periods; the prime and inlet valves are normally open and the isolation and dump valves normally closed, so the driver retains mechanical braking if the vehicle loses power.
- **Skills:** MATLAB / Simulink, TI C2000 F28069 DSP, PMSM, dq-frame current loop PI tuning, CAN, system identification, recursive least squares, solenoid valve control and PWM, dSPACE MicroBox, load cell and pressure sensor measurement

EDUCATION

National Taipei University of Technology M.S., Institute of Vehicle Engineering, Control Group (early admission in the fourth undergraduate year); thesis on parameter estimation, road surface identification and fuzzy control for motorcycle ABS 2017 ~ 2019

- Took on 7 industry and research-institute contract projects while studying, including the two ITRI research contracts behind the brake-by-wire project above (NT$700,000 and NT$600,000, roughly USD 22K and USD 19K).

National Taipei University of Technology B.S., Department of Vehicle Engineering 2013 ~ 2017

INVENTION PATENTS (4 filed, 3 granted; 1 as sole inventor, 1 as first inventor among 6)

TW202426860A	Method for Vehicle Driving Route Planning (MobileDrive, filed 2022.12, sole inventor, under examination)
TWI719598B	Method for Estimating Vehicle Road Friction Coefficient (NCSIST, granted 2021.02, first inventor among 6)
TWI722875B	Control Method for Motorcycle Anti-lock Braking System (National Taipei University of Technology, granted 2021.03)
TWI655435B	Vehicle Speed Estimation Device and Method (HAITEC, granted 2019.04)

ADDITIONAL INFORMATION

Portfolio	KanMingKai.github.io: technical decisions and architecture write-ups for the brake-by-wire, LFS, TRJ and AMR systems above
Languages	Chinese (native), English (technical reading and writing, meetings with international suppliers and OEM customers)